REAL DISUNITY / ENGINEERING DESIGN

Unity-shaped compatibility kit

Engine integration and test plan

Unchanged C# source
Public facade contracts

↔

Rust simulation
Shared host and renderer

Architecture, case catalog and Snake fixture specimen

DESIGN REVISION 1 / 20 OCTOBER 2026

54 proposed case definitions: 50 compatibility-facing and 4 harness engineering.
Runtime implementation and game execution remain planned.

REPORT NAVIGATION

Integration plan and contract reference

The architecture follows the developer workflow from engine ownership and managed dispatch to execution, game admission and reporting. Appendices retain ABI layouts, profile dimensions and the fixture schema.

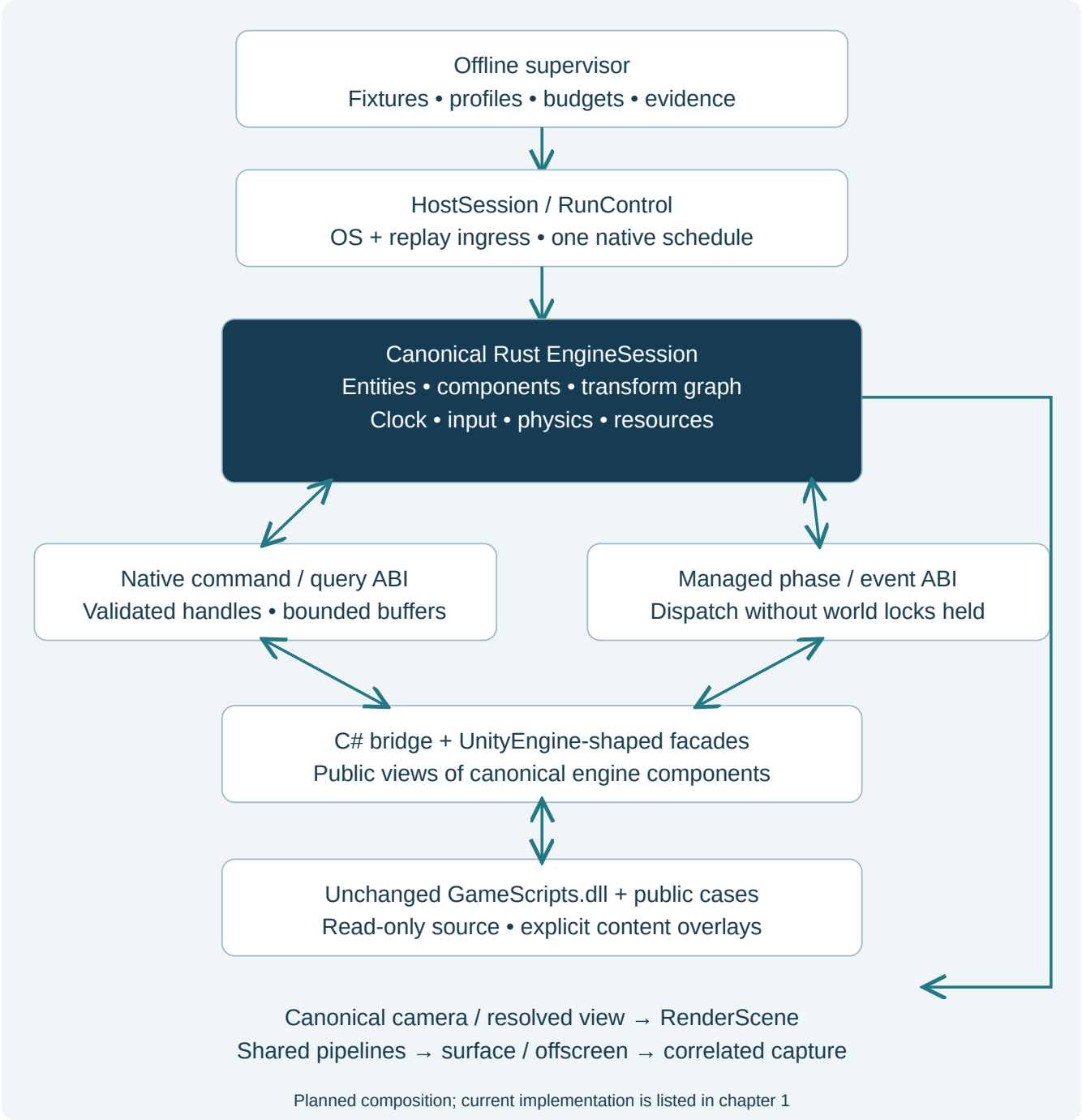
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Diagrams and implementation backlog

Engine and bridge boundaries
Replay and capture sequence
Dashboard layout
54-case catalog
Snake specimen and recorded hashes
Local source references
Build and validation

PLANNED ENGINE COMPOSITION

Native state and managed dispatch



Commands, queries and callbacks reach the same EngineSession. The native schedule releases locks before managed callbacks. Render snapshots and the resolved camera feed the shared surface/offscreen path.

WORKER EXECUTION SEQUENCE

Input, gameplay observations and capture



Simulation time, frame/fixed-tick stamps and wall deadlines have separate roles. Synthetic replay and OS input delivery retain their own evidence labels.

DASHBOARD LAYOUT SPECIFICATION

Profiles, prerequisites and case results

REAL DISUNITY / COMPATIBILITY KIT

DESIGN PREVIEW · EXECUTION PENDING

Run: unavailable Engine revision: unselected Gate: unselected

Source: 6000.3 / 2017.1 | Package: separate selection | Layer / OS / backend

54
Proposed case definitions

50 + 4
Compatibility-facing + harness

Pending
Pass/fail totals require execution

Subsystem × execution profile

Subsystem	Modern / simulation	Modern / desktop	Legacy / desktop
Objects / lifecycle	Unmeasured	Unmeasured	Unmeasured
Input / clock	Unmeasured	Unmeasured	Unmeasured
Camera / sprites	Unmeasured	Unmeasured	Unmeasured
Physics / UI / assets	Unmeasured	Unmeasured	Unmeasured

Snake

25 source requirements · modern

Original sprite missing. A substitute-content variant needs its own asset digest and label.

Breakout

43 source requirements · modern

Physics, tags, stale fields and selected URP or translated-unlit profile require implementation decisions.

Historical Pong

27 source requirements · 2017.1

Legacy prefab/events, score UI and body reset need integration. Eleven historical tests form a separate population.

rd-stranded

Exploration lane · migration

Migrate host, input, camera and capture while preserving game-specific content and render capabilities.

The separate traceability panel tracks 95 gameplay/support requirement instances. Each result links its expectation revision, prerequisite chain, public observations and artifacts. A playable label requires completion of the fixture’s declared journey.

ARCHITECTURE

Unity-shaped compatibility kit: engine integration and test plan

Real Disunity · design revision 1 · 20 October 2026

Engine developers need to know which public contracts work, what prevents a game from running, and whether an engine change breaks an established game journey. This kit will compile unchanged C# game scripts against independently implemented UnityEngine-shaped assemblies, execute them through the Rust engine, and retain assertions, logs and captured frames for each selected contract.

The implementation sequence is Snake, Breakout, historical Pong, then rd-stranded host convergence. Each game adds a bounded set of engine and content requirements. The case catalog defines the initial 54 cases, and the Snake fixture specimen shows the input, replay and evidence contract. The PDF and HTML combine these sources with diagrams and a dashboard layout.

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1. Developer workflow and delivery state

A developer selects a case or fixture, its source and package profiles, an execution mode and a gate policy. Preflight validates the input lock and required capabilities. A worker loads the scene, advances the engine timeline, supplies input and collects public observations. The result records expected and observed values alongside artifacts and teardown status. The dashboard then exposes the dependency that blocked execution or the assertion that failed.

For example, Snake's original-content fixture currently stops at its missing sprite prerequisite. An explicit substitute-content variant could later exercise steering and growth while retaining its own asset digest and fidelity label. Both results would preserve the original script bytes. The original-content result remains blocked until its asset closure is complete.

The first usable release must compile public declaration probes and unchanged Snake scripts in independent assemblies; execute object, transform, lifecycle, clock and input assertions against native state; import a declared content subset; and complete a finite desktop journey with control, gameplay action, reload and renewed control. It must also retain partial results and distinguish assertion failures from infrastructure failures. Adding a game should require fixture data, profiles and, where necessary, an external driver over the shared test protocol.

Current implementation and gaps

This delivery consists of design sources and generated presentation files. The executable kit, UnityEngine-shaped assemblies, native scene importer and game drivers remain planned. The repository already contains a CoreCLR host, a native managed-object store, resource admission, camera controllers and a desktop scene renderer. The following local sources define the starting point.

Capability	Existing implementation	Integration work
Rust application API	worldrunner facade exposes feature-gated gfx, scene, world, script and asset modules.	Add general host/runtime services through this facade so games can use the supported application API.
Content admission	world-model holds semantic documents; world-runtime provides registry/planner, prepared contributions and verified publication.	Preserve exact-schema, capability and content-address checks; materialize admitted contributions into live simulation state.
Asset resources	asset-id, catalog and package provide identities, dependency/variant resolution and bounded packages. Texture, mesh and material define artifacts.	Map imported GUID/fileID/subasset references to these canonical identities and artifacts.
Native object state	disunity-ffi owns generation-checked world/object slots, position, Y yaw, material slot and queued commands under STORE.	Extract/generalize this state into the shared simulation service and connect it to admitted content and rendering.
Managed execution	DisunityHost and ManagedBridge boot CoreCLR, select a bridge, export objects and unload the slice.	Add live phase/event dispatch and script binding.
Managed API	WorldRunner, GameObject, MonoBehaviour and Time provide the current Disunity namespace.	Preserve existing consumers and add independently defined UnityEngine-shaped contracts.
Cameras	camera supplies perspective FreeFly/RTS controllers; SceneView uses a separate governed-scene camera representation.	Introduce a shared resolved-view boundary, arbitrary orientation and an orthographic lens.
Rendering and desktop	RenderScene, SceneGpu, ScenePipelines and OffscreenTarget supply rendering/readback. The viewer app supplies a winit/wgpu surface and resident-scene interaction.	Extract reusable host services; add sorted alpha sprites, UI and gameplay input with presentation-correlated capture.
Observation conventions	ScriptHost and WorldView provide sandboxed Lua hooks, records, seed/budget controls and read-only handles.	Reuse lifetime/observation conventions when designing managed services.
Verification	ABI layouts, store tests, managed integration and quality gates cover existing facilities.	Extend these checks with public managed contracts, importer integration and desktop journeys.

General gameplay input, simulation time, entity/component attachment and Physics2D integration still require implementation. The current native step drains commands, and the managed runner supplies its own narrow callback sequence. Camera representations differ in up-axis and projection support. The current renderer's opaque/masked object pipeline needs alpha ordering and sprite/UI contributions.

Initial scope covers selected source contracts and content schemas on admitted platforms. Full Editor tooling, a Unity package implementation distribution, native plugin emulation, full URP/HDRP, exact shader/solver equivalence and arbitrary precompiled Unity binaries are outside that scope. Physics3D and broader guide

topics are expansion work. Desktop profiles beyond the first Linux X11/Vulkan target require their own host/driver integration.

Expected behavior and the reference guide

Agents wrote the training-only reference from existing knowledge; reviewers corrected it, and pinned third-party games were inspected to compare practical usage with the guide. That creation, review and comparison process used no official Unity documentation or source. Its final summary records 18 groups, 108 topics, 3 errors, 25 omissions, 11 clarifications and 57 remaining questions.

The game comparison records 95 gameplay/support requirement instances: 25 Snake, 43 Breakout and 27 Pong. Five Pong test-tooling requirements form a separate population. The guide-treatment labels are 57 covered, 24 partial, 6 absent and 8 ambiguous. These counts describe written traceability. Execution results use the catalog's separate population of 50 compatibility-facing and 4 harness engineering definitions.

Guide propositions become versioned **expected** records; engine runs supply **observed** records. Signature, phase and format decisions must be settled before their cases enter the required CI baseline. Pending contracts produce inconclusive attempts. The report keeps requirement, case and assertion counts separate and defines no global API-coverage percentage.

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2. Engine integration and state ownership

Unchanged scripts must observe the same transforms and bodies that native simulation and rendering use. The design therefore assigns live simulation state and scheduling to one Rust EngineSession. C# facade objects expose public names and profile semantics over that state. Rust clients, managed scripts and test drivers all reach the same session.

```
Supervisor: fixture + profiles + case selection + gate policy
  -> worker HostSession / RunControl
     -> normalized OS/replay input
     -> EngineSession: entities, components, hierarchy, clock, physics
        -> managed phase/event dispatch -> GameScripts.dll
        <- facade commands/queries through native ABI
     -> immutable render snapshot + resolved camera
     -> RenderScene -> SceneGpu / shared pipelines -> surface or offscreen target
     -> public observations, correlated captures, logs and artifact index

Content: source + explicit sidecar -> versioned importer
         -> existing resource admission -> EngineSession entities/resources
```

These are planned boundaries; package names will be selected during implementation. EngineSession owns live components, resources and advancement. The ABI gateway validates handles, thread entry and buffers. The managed bridge owns assemblies, script instances and cached message binding. Facades own declarations, marshalling and source-coordinate/profile adaptation. Importers own serialized schema and reference translation. HostSession owns window events, input ingress, presentation and capture. The worker supplies assertions and finite plans; the supervisor owns isolation, budgets, aggregation and gate evaluation.

The first structural change is to move the existing FFI store into a CPU simulation service shared by native and managed consumers. Preserve its valid generation/command behavior while migrating the data into that owner. Immutable semantic documents remain admission inputs. Their prepared contributions create live

components in EngineSession. A table/component registry is sufficient for the first increment; storage-library selection can follow demonstrated needs.

The layer registry currently classifies FFI as Foundation and the managed host as Managed, with dependencies restricted accordingly. A generic CPU owner can sit in Foundation; a World-layer materializer can depend on semantic admission. Any required policy change needs an explicit registry update and dependency tests. GPU work stays above the ABI layer, and engine modules remain independent of game and test-driver packages.

Components, resources and visibility

The initial component set is transform hierarchy/TRS, object metadata/activity, managed-script binding, camera/lens, sprite renderer, Rigidbody2D/collider shapes and selected UI layout/text. Resources include the clock/accumulator, normalized device stream, phase input snapshots, seeded randomness, tag/layer registry, asset handles, deferred destruction and physics events.

A facade wrapper is a session-scoped view of a native component. Immediate setter/getter behavior and callback observations require a coherent read model. A native phase command overlay can provide read-after-write visibility and merge at commit; physics updates then refresh that same model. Rendering extracts immutable phase-stamped snapshots. Each operation's visibility and failure behavior belongs in its contract, including whether earlier commands remain applied after a later error.

Source aliases share engine semantics. Historical `Rigidbody2D.velocity` and modern `linearVelocity` access the same native velocity component. Legacy and package input views share device ingress while retaining their selected edge, axis and update-mode policies.

Physics2D needs one native solver behind the body/shape/query service. Selection must evaluate kinematic MovePosition, sleep/wake, elastic contacts, CCD, filters, shape updates and replay on admitted platforms. Solver caches require an explicit synchronization/ownership policy with canonical components. Record the solver revision and tolerances in profiles. Physics3D remains a separately admitted capability within the same world and schedule.

Shared camera and rendering path

Camera projection must agree across public screen conversions, culling, rendering and screenshots. Extend camera with pose, lens and resolved-view contracts covering perspective/orthographic projection, viewport and clip/depth conventions. FreeFly, RTS and Orbit controllers update that contract. SceneView consumes the resolved view while retaining its governed-scene constructor for existing callers. Explicit orientation/up basis supports valid planar views.

A Unity-shaped Camera wraps the canonical camera component. `orthographicSize` means half the vertical span; Snake's fixture config records 11. Test landmark screen positions, inverse conversions and clipping through the actual shared projection.

The existing governed scene uses Z-up, authored integral millimetres and metre rendering, while the camera controller API uses Y-up conventions. Unity-facing content enters through one declared adapter mapping. A proper rotation such as `(x,y,z) -> (x,-z,y)` converts the up-axis; the profile separately resolves forward direction, handedness and winding. Store the mapping and inverse, and apply them consistently to transforms, normals, gravity, rays and the declared 2D physics plane. Screen origin/depth adaptation belongs at presentation. Landmark and round-trip tests cover the complete mapping.

Sprite extraction adds rect/UV, pivot, pixels-per-unit, tint, sorting and alpha blending to canonical render contributions. UI text uses the same renderer and presentation path. Keep existing texture/material resources, GPU context, pipelines and readback limits. Explicit material translations carry a fidelity label, so an unlit profile and a selected URP2D profile retain separate results.

ARCHITECTURE

3. Managed compilation and phase dispatch

The bridge must support scripts compiled in an external assembly and execute their messages at engine-owned phase boundaries. The current host already supplies the CoreCLR loading mechanism. The work is to extend its contract and lifetime handling while keeping existing Disunity consumers compatible.

Existing host and ABI

DisunityHost::boot checks the selected FFI image, assembly, runtimeconfig and bridge type, claims a single active host and loads the cdylib through libloading. It avoids linking the FFI rlib so the process uses the selected native store. ManagedBridge calls nethost load_hostfxr, initialize_for_runtime_config and the assembly delegate loader. A OnceLock retains hostfxr for CoreCLR's process lifetime.

The native-to-managed entries are ConfigureNativeLibrary, Run, WorldHandle, Unload, TokenAlloc, TokenFree and TokenLive. Rust currently binds extern system callbacks; the fixture uses default unmanaged calling convention. ConfigureNativeLibrary copies the selected NUL-terminated UTF-8 path, and the C# resolver binds imports from the Disunity assembly to that exact image. Its path is fixed in-process.

Managed-to-native calls use generated Cdecl DllImport bindings with exact spelling. The ABI manifest defines 12 exports, and the layout tests fix the current 64-bit result/command/state sizes at 16/24/32 bytes. Appendix A records the layouts, queue limits and current hardening gaps.

WorldRunner.Step submits pending commands, calls native step with frame argument 1, updates managed time, then invokes Start/Update/LateUpdate. Native step ignores dt/frame and drains the queue. Callback writes wait for a later step; Transform getters use managed cache. AddComponent(T) creates an object per behavior. The current base requires virtual lifecycle methods and a private-protected constructor. External Unity-shaped scripts therefore need new component attachment and message-binding contracts.

Independent facade assemblies and unchanged source

Keep Disunity.Engine available for its consumers. Build selected CoreModule, physics, input and UI-shaped assemblies with exact identities recorded in each source profile. If a legacy UnityEngine aggregator is needed, type forwarding must retain one defining identity for each public type. Inventory probes pin actual declarations and assembly identities.

MSBuild compiles original files through Compile Include from a read-only root or byte-identical stage into GameScripts.dll. The lock records compiler SDK, language version, net8.0 TFM, symbols, references, asmdef graph and output hashes. The planner resolves platform rules, version defines and cycles, and separates Editor/test assemblies from player references. Historical source uses .NET 8/CoreCLR with a historical API profile.

The new MonoBehaviour base permits derivation from an independent assembly. Binding scans allowed game assemblies once, resolves exact public/private instance message signatures and inheritance policy, and

caches invocation metadata/delegates. Snake's private Start/Update are examples. Required compile-time attributes include Header, Range, SerializeField and CreateAssetMenu. Editor menu execution remains outside the initial runtime scope.

Use an allowlisted offline dependency graph and independently implemented package-facing declarations. Package manifests select requirements. Snake's installed Input System/2D/test tooling is metadata when selected gameplay uses legacy input. Pong's NUnit, TestTools, Editor and NSubstitute dependencies belong to a separate test profile with explicit graph exclusions from gameplay compilation.

Bidirectional phase contract

Append a handshake covering requested/supported capabilities, native ABI revision, bridge protocol revision and struct sizes. Reject incompatible layouts before loading scripts. Existing v1 exports continue forwarding into canonical state. Host-created session attachment must bind C# to that same session.

Proposed bridge families are DescribeBridge, LoadScripts, DispatchPhase, DeliverEvents, CollectObservations and UnloadScripts. They use paired explicit Cdecl conventions on Rust and UnmanagedCallersOnly C# entries. A version/byte-size-tagged FrameContext contains fixed-width session/frame/fixed-tick IDs, phase, rational simulation time, projected dynamic/fixed/scaled/unscaled deltas and flags. Callback records contain binding handle/generation, message kind and bounded payload offsets/lengths. Freeze offsets/alignment with a C#/Rust round-trip test before implementing providers.

The call flow releases native locks before entering managed code, because script callbacks may immediately P/Invoke back into the session:

```
Managed write:
GameScripts -> Transform.position setter -> source-space marshalling
 -> native command/query ABI -> EngineSession Transform
 -> coherent read at the operation's declared visibility point

Engine phase:
HostSession -> EngineSession.begin_phase -> build callback/event batch
 -> release mutable world borrow and store lock
 -> ManagedBridge.DispatchPhase(session, frameContext, callbacks)
 -> scripts query/submit through the native ABI
 -> collect structured callback outcomes; end managed invocation scope
 -> commit allowed commands and continue the native schedule
```

Dispatch comes from the native schedule. Synchronous creation/activation can nest approved initialization messages through a reentrancy-safe boundary after locks are released. Recursive Advance is rejected deterministically.

Native owns input batch memory for the call; managed code copies retained data. Commands, observations and diagnostics use bounded caller buffers. CLR object pointers, Rust references, MethodInfo and managed exceptions stay on their respective sides of the boundary. New UTF-8 strings use pointer+byte-length. ABI, bridge and source-profile revisions remain independently versioned.

Guard load/dispatch/unload entries and catch individual callback exceptions. Record managed type/message/stack and phase, then apply an explicit policy for unrelated eligible callbacks. A case can expect that diagnostic or error when progress becomes impossible. Contain native panics at C entry where unwinding is available; the supervisor records abort/crash as infrastructure error. Terminate only the worker's owned process group on deadline.

Identity, lifetime and thread contract

Identity must survive slot reuse, unload and multiple worlds. Wrapper keys contain session/world generation, object slot/generation, component slot/generation and canonical type ID. Type mapping uses assembly identity, full name and schema/profile revision; importer metadata maps script GUID identity. Retire slots on generation overflow and invalidate all affected wrappers on destroy/unload.

Selected Object equality/bool operators expose destroyed liveness while the CLR reference remains. ReferenceEquals, is null and null-conditional operators retain CLR behavior. Deferred Destroy follows the chosen lifecycle boundary. Existing Disunity IDisposable behavior can remain compatible alongside that facade policy.

EngineSession owns native components/assets. Script bindings root managed instances until destroy/unload and release root tokens during teardown. Finalizers cannot call main-thread APIs. Disposal is idempotent. A fresh process resets static game fields and CoreCLR state; collectible AssemblyLoadContext support is later work with shared facades retained in the default context.

Bind sessions to the engine thread and reject unsupported-thread object queries/mutations with structured errors. Render threads consume immutable snapshots, and asset workers return prepared data for main-thread commit. Pure math and driver computation can run independently. Retained delegates need explicit rooting/unregister lifetimes; caller buffers expire when a call returns.

ARCHITECTURE

4. Test execution and evidence

A declaration probe answers whether selected source forms compile. A runtime case answers whether public behavior matches its expectation. A content case adds import/reference fidelity, and a desktop journey adds actual control and visible gameplay. Keeping these layers separate lets a developer locate the next missing capability.

Layer	Entry point and evidence	Established contract
Inventory	Compile public snippets and inspect independent assembly metadata.	Selected signatures, members and assembly identity.
Semantic	Public managed calls and callback records over the engine timeline.	Bounded native-backed behavior.
Content	Import pinned scenes/assets; inspect public references and rendered output.	Selected schema, prefab and resource fidelity.
Desktop	Shared host/driver, OS input, presentation, public state and reset.	Named game journey on a recorded desktop/backend.

Use real deterministic collaborators for domain and subsystem tests. Public managed probes may read selected public members through a test-owned reflection adapter. Pass criteria use supported observations, actual contacts and rendered output. Private dumps remain failure diagnostics, and drivers cannot invoke game collision handlers or mutate private gameplay fields. Static dependency/layout checks are recorded as engineering checks.

Each case has a stable caseId and separately versioned contractRevision, priority, layer, subsystem, public entry points, inputs/observations, applicability, prerequisites, seed/timeline, tolerances, traceability and artifact policy. Subordinate assertions also have stable IDs. The catalog is the planning authority until executable contracts can generate it. Past observations retain their original expectation revision.

Open destruction/query order, creation-phase eligibility, alias metadata and solver behavior need bounded decisions before gating. Assert the required partial lifecycle order and leave peer order unconstrained. Tolerances name the quantity and operation; physics cases record solver/backend settings and contact/position windows.

Runtime schedule

Normal play and tests advance through the same engine schedule. Proposed observation boundaries are:

1. Ingress timestamped device events and platform changes; update the selected legacy/package snapshots.
2. Accumulate scaled time and run zero or more fixed iterations. Ensure per-instance Start eligibility before its first selected update, dispatch FixedUpdate, commit body/shape writes, step physics, publish contacts and dispatch selected messages with locks released.
3. Ensure Start for newly eligible dynamic participants, dispatch Update and dynamic coroutine continuations, then commit coherent writes.
4. Dispatch LateUpdate; apply the selected destruction boundary; extract transforms/render contributions and settle UI layout.
5. Render/present when requested, capture the tagged frame, run selected end-of-frame continuations and publish observations.

These are proposed engine boundaries. Each timing case pins exact sampling and fixed-versus-dynamic first-participation rules before promotion. Activation can synchronously initialize new objects. Managed coroutine enumerators resume from this schedule.

Represent deterministic time as rational values, with integer-nanosecond stamps where needed and recorded float API projections. A 1/60 dynamic step and .02 fixed step then produce five fixed iterations over six frames without accumulating rounded-nanosecond error. A rendered frame may contain zero or several fixed steps. Pause retains dynamic/unscaled progress while scaled fixed work pauses. Clamp/catch-up policy is separate from Snake's own one-tick-per-Update accumulator.

RunControl and worker lifecycle

RunControl/1 is a proposed engine/test-owned API, usable in-process or through a bounded local pipe. It gives every fixture the same launch, timeline and teardown operations.

Operation	Result
Prepare(runSpec)	Admission, closure, profile and prerequisite report; window creation waits for success.
Load(entryScene)	Session identity and scene-ready event, or structured error.
QueueInput(events)	Accepted sequence IDs or validation diagnostics.
AdvanceTo(point)	Phase/frame/fixed-tick stamps for deterministic progress.

Operation	Result
Observe(probeIds)	Typed public values, sample phase and units.
Capture(request)	Requested/submitted/presented frame IDs and artifact, or error.
Reload / Stop(reason)	Bounded reset/teardown and resource-drain report.

Serialize commands per session. QueueInput rejects past or non-monotonic events unless declared same-timestamp sequencing resolves order. Prepare/Load failure leaves no half-loaded session. Supervisor disconnection cancels the worker, and wall deadlines remain independent of simulation time. Finalize an artifact index after assertion failure; synthesize an error record when a crash leaves an incomplete stream.

Stage hooks such as scene-loaded, after-physics, after-update, render-submitted and before-unload belong to engine/test plumbing. A game-specific driver runs externally or in a test assembly over RunControl and public observations. Original game scripts remain unchanged.

Input and execution modes

A normalized event carries sequence, run-relative time, device ID, kind, control, value and origin. Initial kinds cover device add/remove, key down/up, pointer motion/button/wheel and focus. Preserve physical key/text distinctions, repeat flags and event order.

OS translation and replay injection meet before derived input updates. Legacy down/up edges remain stable for every poll in a dynamic phase; package controls obey the selected dynamic/fixed/manual mode; axes use imported settings. Short taps between fixed ticks remain observable at the appropriate dynamic phase. Focus loss reconciles held keys through the same service.

Normalized replay and OS delivery have separate checks. The first OS driver targets an isolated Linux X11 display, following the viewer smoke workflow. Windows, macOS and Wayland need admitted drivers. Workers use only an explicitly selected display/test context.

Mode	Required resources	Evidence
simulation	Controlled timeline; CoreCLR for C#; CPU only.	Object, lifecycle, input and physics state.
visual-offscreen	Same timeline; admitted GPU/software adapter and OffscreenTarget.	Actual geometry/material/pixel output.
desktop-replay	Main-thread event loop; isolated/explicit display; advancement gated by render/capture completion.	OS/shared-ingress control, presented frames, captures and gameplay state.
interactive	Real clock/input; bounded recorded exploration.	Observed control/action with matching artifacts and a replay plan where feasible.

Reuse the viewer's surface format selection, Fifo presentation and resize/lost/outdated recovery. Suspend, zero viewport and occlusion cannot count as presented frames. Classify an unmet presentation deadline by cause. Initial desktop CI targets Linux X11/Vulkan with an admitted lavapipe adapter. Record GPU and wall timing per environment.

Capture the render target associated with the snapshot sent to presentation; selected desktop checks also capture the OS window. Metadata includes session/frame/tick, requested/submitted/presented IDs, orientation, linear/sRGB policy, viewport, DPI, cropping, view and adapter/driver. OffscreenTarget currently returns top-to-bottom RGBA8 sRGB with bounded readback and a 30-second map deadline. Add frame/session correlation. Prefer geometry/region predicates and declared image tolerances across drivers.

Playable journey acceptance

A playable result requires a recognizable presented scene, input-correlated public state and screen changes, a real gameplay action/success, reset or recovery, and continued control afterwards. Retain logs, screenshots, assertions and a linked exploratory recording/note. A window-open event or static image alone provides only that narrower observation.

Journey names define the claim: Snake food/growth and engine reload; Breakout brick hit/loss/respawn; Pong goal/score/reset/relaunch; rd-stranded scene exploration. The next chapter defines the corresponding fixture boundaries.

ARCHITECTURE

5. Source preservation and game plans

A fixture must identify exactly which scripts, assets and settings produced its result. Keep original inputs read-only and hash every compiler-visible script, metadata/scene/prefab/config/settings file and external dependency used. Recheck the input bytes after execution. Compile original source without source rewriting or IL patches.

Generated build plans, facade references, importer choices, reference fixes, material translations and driver/test assemblies live in overlay/. Each content operation records source object/property, original-value precondition, replacement, rationale and digest. A failed precondition blocks admission. Results identify original-content and unchanged-source/overlaid-content variants separately. Overlays preserve game methods and physics rules; they cannot manufacture supported original art.

The importer walks exact supported text schemas, constructs GUID/fileID/subasset/script identity, validates reachable references and binds fields before activation. Clones remap internal graph references and share external asset references. Diagnose missing edges, duplicate identities, unsupported cycles/structures and mandatory unknown data before partial activation. Preserve stale-field diagnostics until an explicit migration is selected. Converted hand-authored graphs remain useful semantic fixtures with their own format label.

The comparison inventory retains detailed source/requirement evidence. Pins and entry points below come from that inventory; the Snake specimen additionally records selected byte hashes.

Snake: first source-to-desktop slice

Fixture B is eguldali--Optimized-Snake-Game at `0ab43c53e834db714439be71c4151d953d241847`, saved with 6000.3.7f1. Its entry is Assets/Scenes/GameScene.unity with eight C# files. The supplied source root is /tmp/real-disunity-modern-games/eguldali--Optimized-Snake-Game.

Its selected closure needs integer math/operators/hash, private lifecycle binding, serialized config/references, component graph cloning, parent-local placement, pooled activity, legacy W/S/A/D edges and orthographic

sprites. Config is 40×20, wrap enabled, cellSize 1 and eight ticks/second. This slice can establish the runtime/input/render path before adding native physics or URP.

Both visual prefabs reference sprite GUID `311925a002f4447b3a28927169b83ea6`, fileID `7482667652216324306`, with no defining local Assets metadata. The original-content visual lane is blocked. A separately hashed, independently owned substitute can be an explicit bringup variant. The default specimen leaves that variant disabled. Material remaps also need a declared identity and validated translation.

The source has application limitations: GameLoop consumes at most one tick per Update; its model path omits non-wrap bounds death; FoodSpawner can exhaust attempts and return occupied food; tail/growth occupancy has edge cases; visual/config cell sizes are only aligned in the selected configuration. Preserve these behaviors and use the declared wrap/cellSize settings. Reset uses engine scene reload or a fresh worker because the source has no dedicated restart UI.

The driver observes public FoodSpawner.CurrentFoodPos and active SpriteRenderer positions/enabled state under the admitted SnakeView clone parent. Validate selectors against the imported graph. A bounded public-state driver steers toward food, checks turn/wrap, then observes later growth, food relocation and speed manifestation. Characterize a source-supported self-contact/Game Over scenario separately. After reload, check renewed movement. The specimen's fixed W/A sequence demonstrates steering; food routing is a separate driver task.

Breakout: physics and package input

Fixture A is admtrv--Breakout at `2b960a3589f30b6e0025f2509ef00bf9155d6f45`, saved with 6000.3.5f2. Assets/Scenes/SampleScene.unity uses seven C# files. The supplied source root is /tmp/real-disunity-modern-games/admtrv--Breakout.

Breakout adds dynamic/kinematic/static 2D bodies, contacts/triggers, CCD, materials, collider resize/disable, coroutine replacement, spritesheet subassets/PPU and live Tilemap wall collision. Its package profile uses Input System 1.17.0 with A/D fixed polling. Quality tiers select a URP17.3 2D closure despite a zero default pipeline setting.

Content admission needs explicit decisions. Ball prefab uses the Ball tag while project tags are empty; a tag sidecar must be recorded. Serialized startSpeed is stale relative to speed, and brick fields also need diagnostics. Any migration belongs to a preconditioned overlaid-content variant. Original URP2D support requires selected camera/light/renderer/material behavior. A named translated-unlit profile can support a separate gameplay result. Missing sprite-pack content and usage authorization remain prerequisites.

The journey moves/clamps the paddle with A/D, observes real ball contacts and brick animation/collider disable/destruction, then records ball loss and respawn. Seeded public-action cases cover power-ups independently. Settle the pending-Destroy query contract before the last-ball/respawn assertion becomes gating.

Historical Pong: legacy serialization and visible scoring

Fixture C is pong-tdd-licensed-v1 at `e2d3368728e236b0ab5a04afb142a3ae49260dd1`, saved with 2017.1.0f3. Assets/Scenes/Main.unity has 20 gameplay/support C# files; 11 historical tests are a separate population. Source is supplied at /tmp/real-disunity-tck-research/pong-tdd-licensed-v1. The project has no Packages manifest. Seven Resources prefabs support selected test fixtures.

Its separate 2017.1 profile adds legacy aliases/axes, old prefab overrides, persistent enum UnityEvent dispatch, body reset/sleep/elastic response, Canvas/RectTransform/Text and fonts. Goal's serialized team

differs from scoresTo, and Goal+ScoreEvent differs from the current top-level ScoreEvent, with asymmetric scene overrides. A reviewed property/type-map sidecar must specify old values and identities. Ambiguous goal ownership blocks that lane. NSubstitute.dll remains an unresolved test-tooling dependency, admitted separately from gameplay.

W/S and arrow Up/Down feed imported Vertical/AlternateVertical raw axes. The journey observes both paddles, native bounce/goal trigger, persistent listener delivery, public ScoreManager/ScoreView values and numeric text pixels. Reset must zero linear/angular velocity, teleport and sleep the body. After the scaled continuation window, the ball shows, wakes and moves again. Sample after the required continuation opportunity and retain renewed control after reset.

rd-stranded: host convergence and exploration

The previously inspected external checkout at /home/acoliver/projects/worldrunner/branch-2/rd-stranded is outside this workspace. Its games/stranded/Cargo.toml combines engine crates pinned at `0c14471700425aed4012b806f5c48620b02429f3` with renamed local stranded render/host/game/save/world crates. These paths describe migration seams; this design adds no dependency on that checkout.

The recorded games/stranded/src/main.rs creates worldrunner_host::App and supplies profiles, capture commands and native-world paths for a World Explorer. Host app_state.rs pumps winit events. camera_input.rs contains CameraInputState, RtsInputState, CameraBindings and a 1/60 FreeFlyScriptStep replay through real bindings. RenderState owns FreeFly/optional RTS cameras and capture state; camera_matrices.rs and render_state_native.rs contain projection integration to converge with the shared view contract.

The recorded disunity_scene.rs loads a managed slice, exports objects, unloads it, re-grids the snapshot into terrain and builds renderers. lua_shim.rs extracts moon profile data through a selected load-time environment. Those facilities support static content/profile generation. The external Game/GameProfiles composition also carries terrain-specific inputs that should remain at the game edge. These external observations are retained from the earlier inspection and need revalidation against a pinned checkout during migration.

Extract event ingress, timeline, resolved view, presentation/capture and observations into general services. Migrate the current viewer first, then rd-stranded incrementally, preserving its advanced render capabilities through explicit capability migration. The initial journey uses the shared protocol to load an admitted world, move the camera by keyboard/mouse, capture landmarks, reset pose and continue control. Its result is exploration. A full gameplay claim requires a separately inventoried executable content closure.

ARCHITECTURE

6. Results, dashboard and gate policy

A result must let a developer reproduce an observation and identify whether the engine, facade, importer, fixture or platform prevented progress. The proposed real-disunity.tck-run/1 JSON is authoritative. JUnit projects that record into CI's more limited status model.

Run identity includes engine revision/dirty digest, runner build, catalog/expectation hash, profile tuple, source/overlay/assembly hashes and environment/prerequisite report. Record OS/backend/adapter/driver/display/viewport/DPI, seed and schedule. Each case retains selected contract revision, selection reason, attempts, prerequisite dependencies, ordered assertions and expected/observed typed values with units and tolerances. Artifact entries contain relative path, media type, bytes, SHA-256,

origin and session/phase/frame/tick. Final fields include truncation, teardown, gate ID/hash, regressions and exit disposition.

Example record for the blocked original-content Snake fixture:


```
{
  "schema": "real-disunity.tck-case-result/1",
  "runId": "example-not-executed",
  "caseId": "UCK-GAME-001",
  "contractRevision": 1,
  "profileId": "unity-6000.3-source-v1",
  "status": "blocked",
  "attempt": 1,
  "prerequisites": [{"id": "content.snake.sprite", "available": false}],
  "assertions": [],
  "expected": {"journey": "steer, eat, grow, reload, move"},
  "observed": null,
  "diagnostics": [{"code": "missing.asset_reference", "guid": "311925a002f4447b3a28927169b83ea6"}],
  "artifacts": []
}
```


Assertions record assertionId, observationPoint `{frame,fixedTick,phase,time}`, comparator, expected, observed, tolerance, outcome and diagnostic/artifact references. Encode exceptional numeric values explicitly because JSON numbers exclude NaN/Infinity. Missing probes remain missing. Evidence identifies whether it came from a public API, callback, pixel region, platform event or infrastructure diagnostic.

Use schema-versioned NDJSON for input/phase/callback/log/probe streams with monotonic sequence and run/session/case/attempt identity. Rotate within the run-owned directory and mark truncation. Create-new output and atomic index finalization preserve failed attempts. Incomplete streams retain partial artifacts and receive an error record. Game scripts are executable inputs; workers require process isolation, least privileges and no required network access.

Status contract

Status	Meaning
pass	All required assertions ran and matched, including evidence and teardown.
fail	Completed observations under valid prerequisites contradict an assertion.
unsupported	The selected profile deliberately excludes the capability/version/backend.
not-implemented	An in-scope provider, case or driver remains unbuilt.
blocked	Required external content, configuration, tool or prerequisite is unavailable.
skipped	Explicit selection/policy exclusion with a reason.
error	Engine, compiler or harness infrastructure failed outside a completed assertion.
flaky	Recorded attempts under the same controlled identity have conflicting completed outcomes.

Status	Meaning
inconclusive	Required observations or the approved expectation are insufficient to decide.

A declaration probe rejected by the compiler is fail if its provider exists; an absent provider is not-implemented; a compiler crash is error. Missing display is blocked at preflight, while an admitted surface/device crash is error. A completion deadline is fail when adequate observations show the goal was missed, inconclusive when required evidence is unavailable, and error when the worker hangs/crashes. A missing-reference microcase can pass by observing its expected diagnostic while the dependent positive game lane remains blocked.

reasonCode, stage, knownIssueId and blameDomain are orthogonal fields. Known failures retain fail with their issue reference. Unmeasured is the dashboard state for an absent result record. Retain every flaky attempt.

JUnit maps fail to failure and error to error; unsupported, not-implemented, blocked, skipped and inconclusive become skipped with original status properties. Flaky maps to failure unless an explicit policy permits it. Gates read JSON so the skipped aggregate cannot conceal a required result.

CLI and baseline promotion

The CLI below is proposed. Each output directory is new.

```
cargo xtask tck inventory --profile <profile> --output <new-directory>
cargo xtask tck run --suite micro --profile <profile> --mode simulation --gate harness-smoke-v1 --
output <new-directory>
cargo xtask tck run --fixture <manifest> --mode desktop-replay --gate baseline-v1 --output <new-
directory>
cargo xtask tck report --runs <run-index> --output <new-directory>
```

Exit 0 means completion with the selected gate satisfied. Exit 1 covers assertion/gating regression, a disallowed mandatory status or required-case disappearance. Exit 2 covers invalid invocation/spec/profile/lock. Exit 3 covers execution/infrastructure error, including missing infrastructure in a required preflight lane even when dependent cases are blocked. Exit 4 covers cancellation/deadline without trustworthy completion. Combined categories use precedence 4, 3, 2, 1 and retain all details in JSON.

An explicitly selected, recorded report-only policy may return 0 while reporting fail/unsupported counts. That disposition reports completion under that policy. Gate definitions record required case/profile/platform selections, allowed statuses, known-issue owner/expiry and allowed deltas from a hashed baseline. Begin with harness self-tests and a small established passing set. Promote stable cases with reviewed expectations and artifacts; reject disappearing selection, new skips/unsupported and unapproved baseline edits.

Dashboard views

The first dashboard is static HTML over result JSON and relative artifacts. A local run index supplies history. Filters select source, package, layer, subsystem and platform before aggregating; cells show selected/total counts and exact statuses. Every cell links to cases, prerequisite chains and expectation revisions.

Game cards display independent compilation, semantic, content, desktop and exploratory evidence, with source/overlay digests and visible historical/substitution labels. A playable label requires the declared journey's acceptance policy. Case detail places expected/observed values and tolerance beside logs, attempts and actual images. A timeline correlates input, fixed phase, contacts, callbacks and presentation using simulation time, frame/tick and wall time.

The traceability panel keeps the 95 game requirements separate from 54 definitions and assertion totals. Trends compare only matching case revision/profile/platform identities; expectation/source changes appear as discontinuities. Before execution, status cells are unmeasured and pass/fail totals are unavailable. Use keyboard-accessible text alongside color, escape artifact paths/content and keep report artifacts inert.

ARCHITECTURE

7. Delivery and verification

Start with contracts and state ownership so later game results have reproducible identity, scheduling and evidence. Snake establishes the first source-to-desktop path without a physics dependency. Breakout adds physics/package/content depth; Pong exercises a separate historical profile; rd-stranded migrates existing host capabilities.

Milestone	Increment	Promotion evidence
M0: contracts and supervisor	Schema/fixture/result/gate validators, status reducer, quotas, compile probes and static report.	Invalid input/drift rejected; controlled child crash/timeout retains error artifacts; 54 definitions reconcile; absent runs remain unmeasured.
M1: native runtime and bridge	Characterize/extract FFI state; add schedule, phase ABI, independent facades, identity, threads, guarded callbacks, clock/input/random and RunControl.	Existing triangle/ABI behavior retained; public object/transform/lifecycle/time/input cases; 100 load/unload cycles drain worlds/tokens; callback reentrancy/lock checks. Add native-body visibility when physics exists.
M2: Snake	Field/reference import, clone/activity, integer APIs, resolved orthographic camera, sprite rendering and desktop driver.	Eight unchanged scripts; imported config; turn, food/growth, reload and renewed control with captures/exploratory record. Admit each content variant separately.
M3: Breakout	Select/integrate Physics2D; tile/spritesheet, package input, coroutines and selected pipeline capabilities.	Seven unchanged scripts; locked tag/stale-field choices; actual brick/loss/respawn; resize/disable/contact checks; separate fidelity results.
M4: Pong	Legacy aliases, serialization/events/axes, UI/text/fonts and body reset/relaunch.	Twenty unchanged gameplay/support scripts; real goal/listener/visible-score/reset/wait/relaunch. Classify eleven historical tests separately.
M5: rd-stranded	Migrate host/input/view/capture; preserve advanced rendering and game-edge content/profile adapters.	Shared protocol/launcher/results with exploration input/capture/reset; compare existing behavior on a pinned checkout.

Implement one public behavior at a time: add a regression test, confirm failure for the intended missing behavior, implement it, then refactor with tests passing. Characterize the current FFI queue/cache/lifetime behavior before extraction. Preserve source-consumer compatibility and register new packages in layer/quality

policy. Cross-game driver contract tests should exercise the same prepare/load/input/observe/reload/stop sequence.

Verification lanes and budgets

The quality guide defines existing commands. Extend them at the boundary that produces each result.

Lane	Existing base and extension	Resources and artifacts
Static/CPU	cargo xtask fmt, policy, source-size, ci-cpu; add schema/lock/profile/gate tests.	Available locked/offline dependencies; diagnostics and status-reducer/catalog checks.
Managed/ABI	ci-managed; add public worker cases and round-trip layouts.	.NET 8, hostfxr/nethost, selected native image; metadata, callback traces and drains. Existing workflows declare Linux/Windows/macOS jobs.
Simulation	New canonical runtime cases.	Controlled seed/clock and bounded process; state, input/events and teardown.
Offscreen GPU	ci-gpu/render tests plus camera/sprite/UI cases.	Admitted adapter/features/driver; PNG, frame metadata, geometry and readback diagnostics.
Linux desktop CI	Extend isolated-X11 viewer smoke to game replay.	Xvfb, verified software Vulkan, platform driver; existing smoke also requires Pillow/xdotool. Retain OS/render captures, state/logs and clean exit.
Hardware/interactive	New admitted profiles, initially Linux.	Explicit display/test context, actual control and exploratory record; device-specific timing.

Initial proposed defaults are one session/window and one managed worker per process; two compile jobs; 10 simulated seconds/60 wall seconds per case unless raised; 1 GiB resident limit where enforceable; 10 MiB logs; 64 captures; 256 MiB per readback target; and 512 MiB artifacts. The readback default comes from OffscreenTarget; the other values are budget proposals. Game fixtures declare overrides: the Snake specimen uses 60 simulated seconds, 120 wall seconds and 16 captures. Report unenforced resource limits explicitly. Use condition-driven bounded waits and distinguish frame schedules from performance thresholds.

Unit tests cover status/gate/profile/lock/selector/timestamp rules. Integration tests use real FFI/CLR, importer graphs, assets, solver and capture. A small E2E set covers each admitted desktop journey. Disposable children exercise crash, hang, lost prerequisites and truncated streams. Promote only stable, artifact-backed cases into the growing baseline.

The main trade-off is extracting current narrow state/host facilities before broad API expansion. It adds initial integration work and makes later games share corrections. Explicit profiles and fail-closed schemas add metadata maintenance while preserving reproducibility. A local static report is sufficient until concurrent producers require centralized indexing. Extend schemas and policies when additional real fixtures demonstrate a need.

ARCHITECTURE

8. Report generation and evidence limits

Run the retained report generator from the repository root to rebuild the HTML/PDF. It reads the architecture, catalog and fixture JSON; checks source-text retention, case identity and presentation integrity; and prints through the existing local Chrome/Poppler tools. The generated report includes its exact input digests and regeneration instructions.

Earlier design checks recorded matching pins/clean working trees for the three supplied games and matching byte lengths/hashes for the specimen's 15 source/content records. Those checks are historical evidence. This prose/layout rewrite preserves that data and does not repeat game-tree inspection. Current presentation checks are reported separately at generation time.

This delivery runs documentation and presentation validation only. Engine builds, managed/GPU suites, executable TCK cases and desktop game journeys remain implementation verification work. The status and milestone sections identify the required evidence before any compatibility or playability result can be published.

ARCHITECTURE

Appendix A. ABI layouts and current hardening work

The current manifest exports version, world create/destroy, object create/destroy, position/yaw setters, command submit, world step, error message, store statistics and bulk object export. build.rs compares generated binding bytes with the checked C# bindings. Keep existing signatures stable, append new exports, regenerate intentionally and update the manifest/version under its policy.

Type	Current 64-bit layout
DisunityResult	16 bytes; alignment 8; i32 code at 0, u64 detail at 8.
DisunityCommand	24 bytes; alignment 8; u64 object at 0, u32 kind at 8, f32 x/y/z at 12/16/20.
DisunityObjectState	32 bytes; alignment 8; u64 object at 0, f32 xyz/yaw at 8/12/16/20, u32 material slot at 24.

abi.rs uses repr(C); generated C# uses StructLayout.Sequential. New APIs use fixed-width fields and explicit packing tests. Marshal enums with declared widths and encode bool/string values explicitly; managed objects and glam types remain internal.

The native store copies command pointers synchronously and rejects more than 1 << 20 commands per submission. Bulk export uses caller capacity/probe/retry. Error copying is byte-counted UTF-8 without a terminator and reports required size on BUDGET. The existing managed helper tries 512 bytes; error detail uses a constant last-error token. Add bounded retry and durable diagnostic identity as needed by the new protocol.

World/object handles use generation in the upper 32 bits and slot in the lower 32. Object bits are scoped by the separately supplied world, so equal bits can occur across worlds. Current gaps include component/type

handles, explicit affinity, generation-wrap handling and total pending-command bounds. Queued application can partially apply before a later command fails. Mutex serialization currently protects storage access without defining engine-thread semantics.

The portable bridge creates three objects and steps once. Its token is a normal GCHandle root allocated for a managed object. Unload releases worlds/tokens; CoreCLR stays resident, and a collectible game-assembly context is still future work. Configure/Run catch exceptions. Run cleanup can itself throw, and Unload/token entries and native panic containment need uniform guards. These are characterization and hardening tasks for M1.

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Appendix B. Profile selection and diagnostics

A source editor label cannot determine package, importer, solver or platform behavior. The run locks these dimensions independently.

Dimension	Selected data
Engine/API	Repository commit, Rust facade revision, native ABI and bridge protocol.
Source API	unity-6000.3-source-v1 for saved 6000.3.5f2/6000.3.7f1; unity-2017.1-source-v1 for Pong.
Public modules	Exact assemblies, types, members, overloads, optional arguments, conversions and aliases.
Packages	Breakout Input System 1.17.0; Snake installed 1.18.0 with legacy gameplay; Pong legacy UI/events.
Import	Scene/prefab/meta revisions, field migrations, texture/sprite settings and stale-field policy.
Rendering	Built-in sprite subset, explicit translated-unlit subset, or selected URP17.3 2D capabilities.
Simulation	Solver revision/tolerances, collision matrix, .02 fixed step, input mode and axis bindings.
Platform	OS/architecture, display/backend/adapter/driver, CPU/GPU mode, DPI and viewport.
Build	SDK/language/TFM, symbols, asmdef/package graph, tooling and native plugin requirements.

Bind cohesive adapters through immutable registries, following the existing runtime registry. Each declares provided members, dependencies, rejected features and applicable case IDs. Common behavior lives in engine components. A demonstrated semantic version difference gets one named policy with paired tests.

Pong requires velocity, FindGameObjectWithTag and historical inherited find spelling. Modern probes require linearVelocity, FindObjectsByType(None) and FindFirstObjectByType. Pin alias metadata in the profile. Exact Instantiate overloads, two-coordinate Vector3 and conversion/operator declarations remain P0 inventory decisions, as recorded in the catalog.

Located diagnostics include unsupported.member, unsupported.pipeline, unsupported.serialized_version, missing.asset_reference, missing.package_contract and plugin.native_unavailable, plus compiler identity/overload errors. Include file/object/member location and dependency chain. Unsupported calls return structured diagnostics and leave the unsupported operation unapplied. Native plugins need an allowlisted ABI/platform contract; otherwise their dependent fixture is blocked while unrelated cases remain selectable.

ARCHITECTURE

Appendix C. Fixture admission contract

The JSON specimen records eight actual Snake script hashes and seven selected content hashes. It is marked designOnly and runnable=false. The script/content records are retained byte-for-byte in this rewrite. Complete input lock, original sprite, compiler/profile locks, selectors and executable drivers remain unresolved.

Block	Admission responsibility
schema/schemaVersion/designOnly/fixtureId/revision	Exact identity and revision; reject a design specimen as a production fixture.
source	Local root binding, repository/pin, saved editor, per-file inventory and complete lock state.
profiles/compilation	Independent dimensions, package-use classification, compiler/reference/symbol locks and output assembly.
entryScene/inputBindings	Selected scene and control-to-action mappings.
prerequisites	Required capability/content/platform value, availability, reason and blocking scope.
overlay/materialRemaps	Sidecar hashes/preconditions, source/replacement identity and fidelity label.
timeline/resources	Finite dynamic/fixed schedule, seed, input times and simulation/wall/resource quotas.
tasks/assertions/screenshots	Public selectors, bounded drivers/completion and meaningful capture purposes.
artifacts/teardown	New-directory outputs, formats, reload/stop and resource-drain expectations.

A production validator requires complete closure and resolved required hashes/selectors, supported capabilities/profiles and an implemented driver. Reject unknown schema fields, path traversal, symlink escapes and undeclared executables/plugins. Task parameters are data for a small driver registry. Admission completes before activation or window creation.

The specimen fixes seed 1701, dynamic 1/60 and fixed 1/50 seconds, 1280×720 at DPI 1, and explicit Vulkan adapter admission. It contains separate launch, OS input, steering, food/growth, self-contact, reset and exploration tasks. Captures record initial, turned, grown and post-reload states. Teardown drains managed roots/native worlds, finishes or cancels readbacks, rechecks original bytes and closes only owned windows/processes.

CASE REFERENCE

Catalog groups and promotion

The 54 definitions below retain stable case IDs, public stimuli, prerequisites and traceability. Pending decisions require an approved expectation before baseline promotion. The architecture defines status and gate policy.

Profiles, evidence and pending decisions

A. Public surface and independent compilation (6 cases)

B. Object identity and lifecycle (10 cases)

C. Math and authoritative transforms (5 cases)

D. Clock, input and coroutines (6 cases)

E. Canonical Physics2D (6 cases)

F. Import, prefab and asset closure (7 cases)

G. Shared camera, rendering and UI (6 cases)

H. Full journeys using the same harness (4 cases)

I. Harness trustworthiness (4 engineering cases)

Totals and baseline promotion

CASE CATALOG

Compatibility-kit case catalog

Design revision 1 · 54 proposed definitions · execution pending

The architecture and test plan explains the shared engine path. This catalog defines the public behavior and evidence needed to implement it. Each case has a stable ID and starts at contract revision 1. P0 establishes the first API/runtime/Snake slice; P1 adds Breakout, historical Pong or further shared-path checks; P2 follows those slices. A versioned CI baseline separately selects required cases.

CASE CATALOG

Profiles, evidence and pending decisions

Modern selects the 6000.3-shaped source profile; Legacy selects the separate 2017.1-shaped Pong profile. Shared cases run under both where applicable. The declarations are candidate contracts derived from the authored guide and local game requirements. Exact metadata and unresolved semantics need a recorded decision before promotion.

Guide links identify the expected-behavior source. A/B/C requirement IDs refer to the comparison inventory: A is Breakout, B is Snake and C is Pong. Ranges are inclusive. Game usage establishes a requirement; an approved expectation defines the assertion.

Declaration cases compile source forms and inspect independent assembly metadata. Behavior cases execute public managed calls against canonical Rust state. Content/visual/game cases add real importer, GPU and desktop resources. The four harness cases establish runner reliability and are reported separately from compatibility-facing results. Pass criteria use public values, callback records and actual contacts/pixels. Private game fields/helpers and configured mock returns cannot supply gameplay success evidence.

Dependencies identify capabilities. A missing in-scope provider is not-implemented, a profile exclusion is unsupported, and missing required content/tooling is blocked. Cases remain selectable during implementation. An absent result displays as unmeasured. A pending expectation produces an inconclusive attempt until its bounded contract is approved. The Snake specimen demonstrates these distinctions for missing content and planned drivers.

CASE CATALOG

A. Public surface and independent compilation (6 cases)

Declaration probes establish the assembly contract before runtime work depends on it. They compile the same source forms used by the selected games.

ID / priority

UCK-INV-001 · P0

Exercise through public boundary
Compile external `GameScripts.dll` deriving `UnityEngine.MonoBehaviour`, with public serialized fields, private `Awake/Start/Update`, Header/Range/SerializeField attributes; compile ScriptableObject + CreateAssetMenu.

Expected evidence and important failure
External scripts derive from the public base with accessible construction and selected attribute properties. Compilation uses public references and ordinary message methods. Runtime dispatch is exercised separately.

Profile / prerequisites
Shared; .NET8 compiler + core facade

Traceability
G01, G02, G18; A-R0001/0019, B-R0001–0003, C-R0001

UCK-INV-002 · P0

Exercise through public boundary
Compile Vector2/3/Vector2Int constructors, mutable lowercase fields, scalar arithmetic, Vector2Int division/equality/hash use, Vector2/3 conversion and `new Vector3(x,y)`.

Expected evidence and important failure
Pin the exact constructors, lowercase mutable fields, optional arguments, conversions and operators in the public catalog. **Pending declaration:** resolve two-coordinate Vector3 and conversion/operator metadata before gating. Compile the recorded source forms unchanged.

Profile / prerequisites
Shared with per-profile declarations; core/math facade

Traceability
G03; A-R0005, B-R0004–0006/0009–0010, C-R0008–0009

UCK-INV-003 · P0

Exercise through public boundary
Compile each game-used Instantiate form: GameObject+position+rotation; +parent; generic Transform+parent; GetComponent/SetActive/Destroy and public convenience properties.

Expected evidence and important failure
Record inferred type, return type and selected overload for each original source form. Pin parent-only overload metadata independently from its runtime placement behavior. Preserve the original Instantiate calls.

Profile / prerequisites
Shared; core facade

Traceability
G01, G04; A-R0003/0015–0016/0022, B-R0017/0019, C-R0002

UCK-INV-004 · P1

Exercise through public boundary
Compile modern linearVelocity and FindObjectsByType/FindFirstObjectByType; legacy velocity and inherited GameObject.FindObjectOfType/FindGameObjectWithTag. Query metadata for defining assemblies.

Expected evidence and important failure
Expose each profile's selected aliases over shared body/query services. Record defining assembly identity and located diagnostics for excluded members. **Pending declaration:** pin exact alias availability in the profile catalog.

Profile / prerequisites
Modern and Legacy separately; core/Physics2D facade

Traceability
G01, G06; A-R0007/0030, C-R0004/0015–0016

UCK-INV-005 · P1

Exercise through public boundary
Compile Keyboard.current/aKey/dKey.isPressed, legacy Input.GetKeyDown/GetAxisRaw, IEnumerator coroutine/WaitForSeconds, UnityEvent<enum>, UI.Text.

Expected evidence and important failure
Compile with explicit package/module references and selected public declarations. Classify missing Input System or UI contracts under the responsible module/profile.

Profile / prerequisites
Package-specific Modern and Legacy; appropriate facade modules

Traceability
G08, G11; A-R0024/0039/0042, B-R0014/0024, C-R0012/0017/0019/0024

UCK-INV-006 · P0

Exercise through public boundary
Compile selected original scripts byte-for-byte under generated external build plan; inspect assembly dependency graph and hashes.

Expected evidence and important failure
With prerequisites admitted, compile 8 Snake, then 7 Breakout and 20 Pong gameplay/support files byte-for-byte. Reject source drift, Editor leakage, duplicate core identity and implementation-package substitution. Report explicit file/assembly exclusions and the separate 11-test Pong population.

Profile / prerequisites
All game profiles; complete source lock + facade closure/compiler/asmdef planner

Traceability
G05, G18; B-R0021/0024–0025, A-R0034/0042, C-R0025; separate C test-tooling population

CASE CATALOG

B. Object identity and lifecycle (10 cases)

Attachment and lifetime must remain coherent across the managed/native boundary. These cases follow public identity and callback evidence through creation, activation and teardown.

ID / priority

UCK-OBJ-001 · P0

Public stimulus
Create GameObject, attach two different test components, query each from object/component and inspect `gameObject/transform`.

Expected observation / negative case
Both attached components expose the same owner and Transform. Public queries retain identity. An absent supported component follows the selected missing-result contract and leaves the object population unchanged.

Prerequisites
Canonical entity/component registry + public facade

Traceability
G01; A-R0001/0003/0022, B-R0001/0017, C-R0001–0002

UCK-OBJ-002 · P0

Public stimulus
Destroy object through public API, advance to declared destruction boundary, create replacement, then access old component/object and compare to null. Repeat across unload/reload.

Expected observation / negative case
Old wrappers stay invalid through slot reuse, world changes and reload. Destroyed facade liveness compares Unity-null while ReferenceEquals(old,null) remains false. Invalid access returns its diagnostic; is null and null-conditional expressions retain CLR behavior.

Prerequisites
Generation-scoped world/object/component handles, destruction + fake-null

Traceability
G01; A-R0004/0013, B-R0017–0018

UCK-OBJ-003 · P1

Public stimulus
Queue Destroy(ball); query FindObjectsByType before/after named phase boundary and record lifecycle/public liveness.

Expected observation / negative case
Record query inclusion, public liveness and deferred destruction at each named boundary. **Pending expectation:** settle pending-Destroy find inclusion/order for the Breakout last-ball branch before promotion, independently of which count permits respawn.

Prerequisites
Query filters, deferred queue, approved phase oracle

Traceability
G01, G02; A-R0013–0014/0030

UCK-OBJ-004 · P0

Public stimulus
Instantiate GameObject prefab graph containing child, two components, internal/external references, with explicit pose and optional parent.

Expected observation / negative case
Clones have distinct graph/component identities. Internal references target clones; external assets stay shared. Initialization observes complete serialized references and the source prefab remains unchanged.

Prerequisites
Clone graph + field/reference binder + lifecycle

Traceability
G01, G05; A-R0015–0017/0034

UCK-OBJ-005 · P0

Public stimulus
Instantiate Transform under parent using exact parent-only call; toggle clone inactive/active and query public hierarchy/pose.

Expected observation / negative case
The returned Transform belongs to the cloned graph and preserves the selected parent-local relationship. Compare it with the world-pose overload. Activation preserves clone identity. Unsupported clone kinds return a diagnostic.

Prerequisites
Clone + shared hierarchy; INV-003 declaration decision

Traceability
G01, G04; B-R0019–0020

UCK-LIFE-001 · P0

Public stimulus
Load independent assembly with private message methods recording public callback events and reading serialized references; advance first and second dynamic frame.

Expected observation / negative case
Observe Awake before Start, Start before the instance's first Update, and LateUpdate after Update. Eligible instances receive Start once. Awake sees complete references. Assert per-instance order while leaving peer order unconstrained.

Prerequisites
Phase bridge + cached private message dispatch + binder

Traceability
G02; A-R0002, B-R0011, C-R0003

UCK-LIFE-002 · P0

Public stimulus
Activate/deactivate parent, child and behavior; advance while inactive, then reactivate pooled graph.

Expected observation / negative case
Public activeSelf/activeInHierarchy and callback records agree with effective hierarchy activity. Inactive objects stop Update participation; reactivation preserves Start-once behavior. Renderer enable changes preserve object lifetime.

Prerequisites
Eligibility/activity hierarchy

Traceability
G01, G02; B-R0017–0018

UCK-LIFE-003 · P1

Public stimulus
From a real managed callback instantiate/destroy/disable another supported object and query new references, then continue timeline.

Expected observation / negative case
Observe immediate initialization visibility and later-phase eligibility under the selected contract. Advancement completes with each eligible Start delivered once and callbacks respecting destruction. **Pending expectation:** pin creation-phase eligibility before gating. Reentrant queries must complete without deadlock or recursive clock advancement.

Prerequisites
Reentrancy-safe dispatch, lock-free callback boundary

Traceability
G01, G02; A-R0017, B-R0018–0020

UCK-LIFE-004 · P0

Public stimulus
Managed Update throws known exception; another eligible component records progress; expected diagnostic is consumed by case.

Expected observation / negative case
Capture managed exception type/message/stack and phase as a structured outcome. Apply the declared unrelated-callback policy and drain all roots/worlds at unload. The case passes when its expected diagnostic and continued progress are observed and the unmanaged boundary remains intact.

Prerequisites
Guarded bridge + callback outcomes + teardown

Traceability
G02, G18; core runtime safety extension

UCK-LIFE-005 · P1

Public stimulus
Attempt object mutation from Task/unsupported worker thread; then mutate on engine thread and unload.

Expected observation / negative case
Reject the off-thread operation with an affinity diagnostic and unchanged state. The engine-thread operation then succeeds and unload completes. Record this API affinity contract independently of storage locking.

Prerequisites
Engine-thread gate + canonical state

Traceability
G01, G17; core public runtime contract

CASE CATALOG

C. Math and authoritative transforms (5 cases)

Grid gameplay and native integration both depend on values that agree through the shared transform graph. Test exact integer behavior separately from floating-point geometry.

ID / priority

UCK-MATH-001 · P0

Public stimulus
Construct/divide/add Vector2Int, compare/hash and insert equal keys into HashSet; include negative coordinates and cardinal vectors.

Expected observation / boundary
Equal value keys hash coherently and deduplicate in HashSet. Arithmetic and grid bounds match exact integer results, including negative coordinates and the selected division truncation policy.

Prerequisites
Public value structs/operators

Traceability
G03; B-R0004–0006

UCK-MATH-002 · P1

Public stimulus
Normalize nonzero/zero vectors, clamp paddle offset, compute sprite rect/PPU/scale spacing and normalized bounce direction.

Expected observation / boundary
Lengths, directions, clamped bounds and sprite spacing match named tolerances. Zero normalization remains finite under an explicit epsilon policy. Limit the claim to that selected policy.

Prerequisites
Vector/Mathf/Rect public implementation

Traceability
G03; A-R0005–0006/0012/0028–0029, C-R0008–0009

UCK-MATH-003 · P0

Public stimulus
Seed run, draw public int Range for small/non-power-of-two bounds and replay; use a separate recorded float Range/value sequence.

Expected observation / boundary
Integer Range excludes max and stays within admitted bounds. Replay reproduces the same engine-policy sequence; float endpoints/range follow the pinned contract. Select degenerate/reversed-range behavior before asserting it. Sequence identity is scoped to the engine random policy.

Prerequisites
Engine random resource + public facade

Traceability
G03; A-R0009–0010, B-R0007–0008, C-R0021–0022

UCK-XFM-001 · P0

Public stimulus
Set local TRS under translated/rotated/scaled parent; read world position; SetParent with worldPositionStays true/false; transform point/direction and inverse.

Expected observation / boundary
Ordinary nonsingular transforms and inverse round trips agree through the canonical hierarchy. Direction excludes scale; point/vector semantics retain their selected differences. Reject cycles without changing the valid graph. Report unsupported shear/singular cases explicitly.

Prerequisites
Canonical hierarchy + coordinate adapter

Traceability
G04; A-R0011, B-R0010/0020, C-R0010

UCK-XFM-002 · P0

Public stimulus
Set position in managed callback, immediately read it; advance shared simulation; change same entity via native test driver/body integration and read managed value next eligible phase.

Expected observation / boundary
Managed reads reflect writes at the declared visibility point and observe subsequent native changes. Render extraction uses the same committed transform. Phase stamps identify any permitted synchronization boundary.

Prerequisites
Sole native state + bridge visibility/read overlay + render snapshot (physics variant P1)

Traceability
G04; A-R0011/0031–0032, B-R0010, C-R0005/0010

CASE CATALOG

D. Clock, input and coroutines (6 cases)

Replay needs phase-stable input and reproducible simulated time. These cases observe the public clock, device projections and coroutine effects at declared boundaries.

ID / priority

UCK-TIME-001 · P0

Public stimulus
Advance six 1/60 dynamic frames with .02 fixed step; callbacks record public dt/frame/fixed observations.

Expected observation / failure
Six dynamic frames total exactly .1 s and yield five fixed steps. Public dynamic/fixed deltas and frame counts match the native schedule. Start eligibility is per instance. Managed callbacks leave clock advancement under native ownership.

Prerequisites
Native rational schedule + public Time + phase dispatch

Traceability
G02; A-R0002/0031/0043, B-R0011–0013, C-R0003/0023

UCK-TIME-002 · P1

Public stimulus
Set selected time scale, advance, pause scaled time then resume; observe scaled/unscaled deltas and fixed participation.

Expected observation / failure
Scaled time and fixed participation follow the selected scale. Pause stops scaled fixed steps/waits while unscaled time progresses monotonically. Record clamp/catch-up limits separately from game accumulator behavior.

Prerequisites
Shared clock + selected Time API

Traceability
G02; B-R0012–0013, A-R0025, C-R0018/0023

UCK-INP-001 · P0

Public stimulus
Inject timestamped key down/up and short between-fixed tap via same ingress used by winit; two components poll GetKeyDown repeatedly in one dynamic phase; include focus loss.

Expected observation / failure
Every poll sees the same phase-stable edge; the next frame clears it. Held state follows events, a between-fixed tap remains observable dynamically, and focus loss reconciles held keys. HAR-003 and game cases separately establish OS delivery.

Prerequisites
Normalized ingress + legacy input projection + phase snapshot

Traceability
G08; B-R0014–0015/0024

UCK-INP-002 · P1

Public stimulus
Deliver held A/D package keys across dynamic/fixed boundaries; separately exercise imported Vertical/AlternateVertical raw axes, releases and unknown axis.

Expected observation / failure
Keyboard current/null and isPressed obey the selected package update mode. Imported Pong W/S and arrow bindings drive the named axes. Unknown axes return the selected diagnostic. Releases update the appropriate phase snapshot.

Prerequisites
Shared ingress, package profile, legacy axis adapter/settings importer

Traceability
G08; A-R0039–0040/0042, C-R0019–0020

UCK-COR-001 · P1

Public stimulus
Start IEnumerator from script, yield WaitForSeconds under known scale, advance to pre-deadline then later eligible continuation point.

Expected observation / failure
The first synchronous segment follows the selected contract. Scaled waits resume at the approved eligible opportunity after their deadline and never early. Observation points constrain the required phase order.

Prerequisites
Native schedule-driven managed coroutine queues

Traceability
G02; A-R0024–0025, C-R0017–0018

UCK-COR-002 · P1

Public stimulus
Start width-like coroutine twice, stop prior handle, later disable/destroy owner; record public effect before/after deadlines.

Expected observation / failure
Stopping the prior owner-bound handle prevents its later reversion of the replacement effect. Destroy/activation follow the selected lifetime policy and release enumerators. **Pending expectation:** pin inactive-owner stop semantics before gating.

Prerequisites
Coroutine identity/cancel + lifetime

Traceability
G02; A-R0024–0025

CASE CATALOG

E. Canonical Physics2D (6 cases)

Breakout and Pong depend on real contact response. These cases exercise canonical bodies, shapes and callbacks with recorded solver/revision/material/fixed settings. Name contact-time windows and position/velocity tolerances for each scenario.

ID / priority

UCK-PHY-001 · P1

Public stimulus
Set modern linearVelocity/legacy velocity through facade, advance fixed steps with zero gravity, inspect position/velocity; include configured gravity variant.

Expected observation / boundary
Native integration updates the canonical transform and body within recorded tolerance, with finite values. Both source aliases expose that same body’s velocity under the selected gravity settings.

Prerequisites
Shared Physics2D service + settings + adapters

Traceability
G06; A-R0007, C-R0004

UCK-PHY-002 · P1

Public stimulus
Real dynamic circle ball contacts kinematic paddle/static wall under elastic material; move paddle through MovePosition from FixedUpdate.

Expected observation / boundary
Actual contacts deliver the other GameObject and selected velocity/contact response within tolerance. Kinematic MovePosition executes through the solver. Pin ball normalization observations before/after physics so script writes and solver response can be distinguished.

Prerequisites
Native bodies/shapes/contact delivery + physical material

Traceability
G06; A-R0008/0020–0023/0031–0032/0036/0043, C-R0006–0007/0026

UCK-PHY-003 · P1

Public stimulus
Move ball into real trigger; filter by imported layer/tag, then disable trigger/body and repeat.

Expected observation / boundary
Entering contact intervals produce the selected enter event with correct Collider2D/owner. Triggers leave physical bounce absent. Disabled/filtered pairs are excluded. A missing Ball tag yields its own diagnostic.

Prerequisites
Contact transitions + tags/layers/filter settings

Traceability
G01, G06; A-R0020/0022–0023/0033/0043, C-R0014/0026

UCK-PHY-004 · P1

Public stimulus
Publicly resize BoxCollider2D and disable a contacted collider at declared boundaries; query/contact before/after synchronization.

Expected observation / boundary
Geometry, mass and contact state update at the declared synchronization boundary. Disabled colliders leave contact participation. Sprite assignment changes rendering independently. **Pending expectation:** pin synchronization timing before gating.

Prerequisites
Shape update/sync policy

Traceability
G04, G06; A-R0027/0036

UCK-PHY-005 · P1

Public stimulus
Launch fast small ball toward thin wall under selected continuous mode and fixed dt, observe actual motion/contact; repeat discrete mode characterization separately.

Expected observation / boundary
Selected continuous mode prevents tunnelling within the bounded scenario/tolerances. Record solver mode/settings and actual contact/motion. Excluded CCD returns a capability diagnostic; characterize discrete mode separately.

Prerequisites
Solver with actual selected CCD capability + settings

Traceability
G06; A-R0023/0036/0043

UCK-PHY-006 · P1

Public stimulus
Set velocity/angular velocity to zero, teleport transform to initial position, Sleep; later assign launch velocity through public alias.

Expected observation / boundary
After a step the body stays stopped at its reset pose and matches the canonical transform. Later public launch velocity wakes and moves it. Renderer toggles preserve body identity. Pin exact synchronization/wake policy.

Prerequisites
Body sleep/wake/transform sync

Traceability
G06; C-R0004–0005/0010–0011

CASE CATALOG

F. Import, prefab and asset closure (7 cases)

Game execution depends on references and serialized values surviving import and clone operations. These cases admit real selected content and locate unresolved edges before activation.

ID / priority

UCK-ASSET-001 · P0

Supported real content stimulus
Import small native text scene + MonoBehaviour + ScriptableObject config with GUID/fileID/script identity and private/public serialized references.

Expected observation / negative case
Content-bound values/references appear in public getters and callback observations before activation. Reject unsupported serialized revisions, duplicate GUIDs and missing fileIDs with location/reference chains. Exercise the native text importer for this case.

Prerequisites
Versioned text importer + binder + script identity map

Traceability
G05; B-R0001–0003/0021, A-R0019/0034

UCK-ASSET-002 · P0

Supported real content stimulus
Load selected scene prefabs, instantiate nested supported graph, honor serialized overrides and cross-component references.

Expected observation / negative case
Public ownership, override and reference values match the selected graph. Clone IDs differ and prefab input stays unchanged. A missing mandatory edge cleans up partial materialization before activation.

Prerequisites
Native prefab schema/profile + canonical clone

Traceability
G05; A-R0017/0034, B-R0019–0021, C-R0025

UCK-ASSET-003 · P0

Supported real content stimulus
Follow missing sprite GUID from Snake visual prefabs; run admitted missing-reference microfixture; then retry a separately labelled explicit remap overlay.

Expected observation / negative case
Diagnose the missing reference and block the original-content visual lane. The negative microfixture passes on its expected diagnostic. The separate remap records precondition/hash/source identity and keeps substitute-content fidelity distinct.

Prerequisites
Closure walker/overlay validator; actual original sprite unavailable for positive Snake branch

Traceability
G05, G10, G14; B-R0022

UCK-ASSET-004 · P1

Supported real content stimulus
Import texture sheet metadata and distinct sprite subassets, rect/pivot/PPU/frame refs, assign through SpriteRenderer.

Expected observation / negative case
Each subasset keeps its own identity, UV region, rect/pivot and PPU. Public sizing agrees with the rendered region. Missing subassets produce located diagnostics.

Prerequisites
Texture decoder + selected sprite metadata schema + renderer

Traceability
G10; A-R0026/0028–0029/0035

UCK-ASSET-005 · P1

Supported real content stimulus
Import rectangular Tile asset/Grid/Tilemap wall and collision settings from selected authored graph; edit supported cell and request declared refresh.

Expected observation / negative case
Visible cells and generated canonical colliders agree with admitted transforms/settings at the declared refresh boundary. Reject unsupported layouts/components. Exercise actual wall contacts as well as rendering.

Prerequisites
Tile importer + canonical tile contribution/shape generator + Physics2D

Traceability
G10; A-R0037

UCK-ASSET-006 · P1

Supported real content stimulus
Bind Pong native score enum persistent listener and old prefab scene overrides using selected legacy schema; read public score after genuine trigger.

Expected observation / negative case
A real trigger dispatches the correct target/method/enum to ScoreManager and ScoreView. Require explicit preconditioned mappings for team/scoresTo and Goal+ScoreEvent, or block ambiguous wiring. Keep binary plugin admission separate.

Prerequisites
Legacy importer + UnityEvent adapter + supported graph/Physics2D

Traceability
G05, G11; C-R0012–0014/0025

UCK-ASSET-007 · P1

Supported real content stimulus
Resolve selected Resources prefab by logical path/type; retain shared resource through clones then unload scene/teardown; inspect missing key.

Expected observation / negative case
Resolution follows the selected path/type/missing-key contract. Clones share the admitted asset and teardown balances native/managed roots. Report import/package/platform/plugin diagnostics separately. Scope this case to the Resources subset.

Prerequisites
Resources adapter + resource ownership + selected legacy assets

Traceability
G14, G18; C-R0025 and separate Pong test fixtures; A-R0038/0042, B-R0025 boundaries

CASE CATALOG

G. Shared camera, rendering and UI (6 cases)

Camera queries and screenshots must agree about the same scene. These cases compare public geometry with actual renderer output, including sprite and score-text changes.

ID / priority

UCK-VIS-001 · P0

Public/visible stimulus
Configure facade camera orthographic size/aspect/pose, project known scene landmarks; render them and resize viewport.

Expected observation / boundary
Half-height, clip/depth and inverse conversions agree with actual landmark pixels after resizing. The declared coordinate mapping preserves right/up orientation and accepts the planar view.

Prerequisites
Shared resolved camera + orthographic lens + capture

Traceability
G04, G09; A-R0041, B-R0023, C-R0027

UCK-VIS-002 · P0

Public/visible stimulus
Render imported sprites of known rect/PPU/pivot/tint and sorting order, toggle active/enabled, update sprite frame.

Expected observation / boundary
Actual target pixels match rect/PPU/pivot, tint, alpha, sorting and hide/show expectations. A frame assignment changes the selected region. Record renderer/owner/collider activity as separate observations.

Prerequisites
Canonical sprite contribution + shared renderer alpha/sorting + admitted assets

Traceability
G09, G10; A-R0026/0028–0029/0041, B-R0017/0023, C-R0011/0027

UCK-VIS-003 · P1

Public/visible stimulus
Admit selected Breakout quality-tier pipeline/material/light asset closure; separately admit explicit unlit-remap profile.

Expected observation / boundary
Require the selected URP17.3 2D behaviors/resources for original-pipeline fidelity. Excluded features return located unsupported diagnostics. Report the explicitly identified unlit-remap variant under its own profile.

Prerequisites
Pipeline capability registry + importer + canonical material/render features

Traceability
G09, G10; A-R0038, B-R0022/0025

UCK-VIS-004 · P1

Public/visible stimulus
Capture tagged rendered frame with known colored landmarks via offscreen and desktop presentation, including resize/zero viewport and readback budget rejection.

Expected observation / boundary
Capture orientation, color policy, dimensions and frame/session identity agree with landmarks. Zero viewport/device failure yields a diagnostic. Check geometry/regions with declared tolerances across drivers and enforce readback budgets.

Prerequisites
Existing OffscreenTarget/shared surface extraction + new frame correlation

Traceability
G09, G15; A-R0041, B-R0023, C-R0027

UCK-UI-001 · P1

Public/visible stimulus
Set Canvas/RectTransform anchored position/pivot/size under known parent resolution, then resize and render score region.

Expected observation / boundary
Public rect values and rendered score-region geometry match selected fixed/stretch relationships after explicit layout settling. Resize preserves the declared relationships through the canonical UI/render contribution.

Prerequisites
Canonical UI layout + selected Canvas subset + GPU

Traceability
G04, G11; C-R0027

UCK-UI-002 · P1

Public/visible stimulus
Set legacy Text.text from public ScoreView path with admitted font; update 0 to 1 and hide/show parent.

Expected observation / boundary
Public Text.text and legible rendered glyphs change from 0 to 1 together and follow parent visibility. Missing fonts block fidelity; an admitted mapped font belongs to an explicitly labelled overlay profile.

Prerequisites
Text/font importer/rasterization + shared UI contribution

Traceability
G11; C-R0024/0027

CASE CATALOG

H. Full journeys using the same harness (4 cases)

Each journey groups stable subordinate assertions under one case ID. Retain separate compile/content/runtime/desktop outcomes, original-source byte hashes, finite driver/time limits and public state with presented/captured evidence. Reset must preserve renewed control, with a linked exploratory artifact.

ID / priority

UCK-GAME-001 · P0

Journey and minimum machine observations

Snake: present GameScene; steer W/A/S/D through shared ingress; correlate visible head/segment positions with control; use public CurrentFoodPos and segment observations to reach food within a bounded driver plan; observe later growth/speed, wrap and a selected self-contact/Game Over scenario; reload or reset the worker, then move again.

Prerequisites and scope

Eight unchanged scripts; admitted 40×20/wrap=true/8-ticks-per-second config; clone/activity/input/camera/sprites. Original sprite is absent. Keep substitute content as a separate variant and use public observations. Reset is engine-owned scene reload or fresh worker.

Traceability

G01–G04/G08–G10; B-R0001–0025

UCK-GAME-002 · P1

Journey and minimum machine observations

Breakout: A/D moves/clamps the visible paddle; real ball contacts lead to brick sprite animation, collider disable and destruction; observe ball loss and respawn, selected seeded power-up/recovery, reload and renewed control.

Prerequisites and scope

Seven unchanged scripts; canonical physics, tiles, materials, input and coroutines. Record tag/stale-field sidecars and separate URP/translated-unlit profiles. Settle pending-Destroy query visibility before gating respawn; the driver supplies public actions through shared ingress.

Traceability

G01–G06/G08–G10; A-R0001–0043

UCK-GAME-003 · P1

Journey and minimum machine observations

Historical Pong: W/S and arrow axes move both paddles; actual bounce/goal contacts dispatch the persistent enum listener; public score and visible text increment; stop/teleport/hide, scaled wait and show/relaunch produce renewed motion. Reset and verify control again.

Prerequisites and scope

Twenty unchanged gameplay/support scripts; legacy prefab, axes, events, UI/font and body behavior. Reviewed sidecars must resolve stale type/field wiring or block the lane. Track eleven historical tests in their separate tooling profile.

Traceability

G01–G06/G08–G11; C-R0001–0027

UCK-GAME-004 · P2

Journey and minimum machine observations

rd-stranded exploration: load pinned native content, present landmarks, drive keyboard/mouse through shared ingress, correlate camera state and pixels, reset pose, continue motion, retain captures/exploration record and tear down through RunControl.

Prerequisites and scope

Requires external checkout migration and a content pin. The current managed slice exports a static snapshot. Initial acceptance covers world exploration; full gameplay needs an inventoried executable closure. Reuse canonical host/render services.

Traceability

Existing external `games/stranded/crates/host` source; shared camera/input/render/run-control contracts

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CASE CATALOG

I. Harness trustworthiness (4 engineering cases)

These four definitions establish status reporting, source integrity, OS delivery and resource lifetime. Include them in the 54-case planning total and report them as harness engineering results, separate from the 50 compatibility-facing definitions.

ID / priority

UCK-HAR-001 · P0

Exercise

Run disposable children that fail an assertion, crash, hang, lack prerequisites or truncate results. Supply explicit profile exclusions and skip decisions, then repeated conflicting completed attempts.

Expected externally meaningful outcome

Record distinct fail/error/blocked/unsupported/skipped/inconclusive/not-implemented statuses and flaky attempts. Bound termination to the child process group, retain or synthesize the artifact index, and check JSON/JUnit plus required-gate exit semantics against the architecture.

Prerequisites

Supervisor/status/gate/report implementation; disposable controlled child fixtures

UCK-HAR-002 · P0

Exercise

Change a staged source byte, overlay precondition or hash; remove a required case; change contract revision; submit unknown schema/profile fields.

Expected externally meaningful outcome

Reject invalid admission/baseline changes while preserving originals and required selection. Correlate source/overlay digests with compiler inputs and keep outputs in new run directories. Verify preservation using byte hashes.

Prerequisites

Fixture/source-lock/schema/baseline validators

UCK-HAR-003 · P0

Exercise

Create/focus an isolated Linux X11 window, send OS key down/up through the admitted driver, observe shared ingress/public callbacks, then resize, unmap/map and close.

Expected externally meaningful outcome

Correlate winit translation, presented capture and public probes. Verify focus release/recovery and exit. Missing display/tool is blocked at preflight; admitted-window crash is error. Require actual OS delivery and separately admitted drivers on other platforms.

Prerequisites

Xvfb/X11 + admitted Vulkan/adapter + driver; generalized host/input; current viewer smoke is foundation

UCK-HAR-004 · P0

Exercise

Repeatedly load/step/export/unload a real managed fixture, initially the existing triangle for 100 cycles. Reload a game session and request render/readback shutdown; observe drains.

Expected externally meaningful outcome

Unload leaves zero live worlds/root tokens and invalidates old wrappers. Native tables remain bounded, and GPU readback completes/cancels before resource release. Use a fresh process for static reset unless a collectible-session contract exists. Retain drain evidence in artifacts.

Prerequisites

Real .NET8/native ABI; shared lifetime/teardown; GPU variant separately admitted

CASE CATALOG

Totals and baseline promotion

Group	Cases	Evidence kind
Public inventory	6	Compile/public metadata
Object and lifecycle	10	Managed public behavior/native engine
Math and transforms	5	Public values/shared state
Clock/input/coroutines	6	Public timing/input/callback observations
Physics2D	6	Actual solver/contact behavior
Content/assets	7	Real import/resource/graph integration
Camera/render/UI	6	Public geometry plus actual GPU/pixels
Game journeys	4	Desktop/state/exploratory evidence
Harness reliability	4	Runner/gate/platform/lifetime correctness
Total	**54**	**50 compatibility-facing + 4 harness engineering cases**

The catalog total counts definitions. Assertion counts, guide topics and per-profile applicable subsets are separate measures. A deliberate diagnostic scenario may pass while the corresponding positive game path remains blocked or unsupported. Every attempt retains expected and observed records.

Promote HAR-001/002 and compilation/lifetime foundations first, then object/math/hierarchy/lifecycle/clock/input, Snake assets/visuals/journey, Breakout physics/package/coroutines, Pong legacy serialization/events/UI, and rd-stranded convergence. For each behavior, confirm a public-boundary test fails for the intended gap before implementation. Baseline changes require reviewed case revision, expectation/profile traceability and stable run artifacts.

Additional guide groups, including 3D physics, animation, audio, broad assets, networking, jobs and Editor tools, remain expansion targets. Selected unsupported or unbuilt capabilities retain explicit status metadata. The initial catalog follows the inventoried game requirements.

FIXTURE SPECIMEN

Snake admission and replay inputs

This non-executable design specimen records source hashes and a planned journey. Its complete lock, original sprite, selectors and implementations remain prerequisites. The presentation build reproduces the recorded data without inspecting the game tree.

Selection	Recorded value
Fixture	snake-modern-original-content
Source pin	0ab43c53e834db714439be71c4151d953d241847
Scene	Assets/Scenes/GameScene.unity
Source API	unity-6000.3-source-v1
ABI / bridge	planned-append-only-extension-of-disunity-v1 / planned-phase-dispatch-v1
Import / rendering	unity-text-scene-prefab-scriptableobject-selected-v1 / canonical-built-in-unlit-sprite-subset-v1
Compilation	.NET 8; independent GameScripts.dll; SDK/language/symbol/assembly locks pending.
Content variant	Original content unresolved; source edits disabled; optional substitute variant disabled.
Input lock	all referenced .meta files; scene/prefab/config reference closure; selected ProjectSettings; built-in resource mappings; overlay and imported resource bytes; compiler-visible inputs and assembly graph

Package metadata

Package	Version	Selected use
com.unity.inputsystem	1.18.0	installed-not-called-by-selected-gameplay
com.unity.feature.2d	2.0.2	editor-tooling-metadata
com.unity.test-framework	1.6.0	not-required-by-selected-gameplay

The manifest excludes Editor/test graphs from gameplay compilation, declares zero native plugins and selects independently implemented facade references. Imported config is {"source": "Assets/GameConfigs/GameConfig.asset", "gridSize": [40, 20], "ticksPerSecond": 8, "wrapAround": true, "cellSize": 1, "speedIncreasePerFood": 0.3, "maxTicksPerSecond": 20, "cameraOrthographicHalfHeight": 11}.

RECORDED INPUT INVENTORY

Script and selected content hashes

The specimen contains eight script records and seven selected content records. These records are part of the future complete-input lock.

Path	Bytes	SHA-256
Assets/Scripts/FoodSpawner.cs	707	`e4f0d72779f914aafd4b1a90789ed102b3c7363da462a1fbcd7bfbe7ab50ffe3`
Assets/Scripts/GameConfig.cs	356	`9c81f1015c95d6fc75987e7ddd3c4dc63bd969be44cc5c5ead188757e8632050`
Assets/Scripts/GameLoop.cs	1552	`ee7e77bff1787c15f089a635307bb143ce1893bf9b8aab2a8a20f68af04c6e9a`
Assets/Scripts/GridSystem.cs	996	`eed2e7d9e492ca99e85761396cc6e1273eee2bbef6e3bd77ab881b98d5ecfff1`
Assets/Scripts/InputBuffer.cs	266	`024c157d3a5c301dde8be0d12c0666e0385d589f0d025ab380fc921be5e28dbd`
Assets/Scripts/PlayerInputReader.cs	422	`ce3b6b920b40afd222ba356a1aff4e9fa04eb59e5b01d62ee213438f82a42094`
Assets/Scripts/SnakeModel.cs	1833	`47d156eae61d8b8ca66a07891dfb5557bbbcbbb0fd33549aec47972d9f3befda`
Assets/Scripts/SnakeView.cs	980	`dde43a55bed814595df956a2f33715cf7dd862a5eeecbc81c99f3a8510cd9ee0`
Assets/Scenes/GameScene.unity		`117e3e4dc56f3e443f984631e63d9da0f9274afdb8a0fc868dafaf12ea62caad`
Assets/GameConfigs/GameConfig.asset		`4af8c14c37b7437446edf1b865d300994c2dc83a6651ee828b1fd34d02b27823`
Assets/Visuals/FoodVisual.prefab		`af0953122bf500f53b48e2d40a7ccf8479e6ed7eb263e19136956f6ccb4956cc`
Assets/Visuals/SneakSegment.prefab		`13938f2bebcfa968a3b6ac84a00b0cd12dda8c1e787bfe9095e70ec54a559c72`
ProjectSettings/ProjectVersion.txt		`50f1bf3fb4196ce6ddf6dbd0eb2ef8c2227a18b5ff762c90e1cb4feab8a06b4a`
Packages/manifest.json		`00a83deb6445f3f3a5740741a6fea4b97ec8c0e05bd23b5a36777bf46bc52fb9`
Packages/packages-lock.json		`d49f8d20afbfbf5891a241dd6334baac84135a978ef8c5f9b644ae3573ac8d44`

FINITE JOURNEY CONTRACT

Tasks, evidence and teardown

Seed 1701; dynamic 1/60 s; fixed 1/50 s; at most 3600 frames or 60 simulated seconds. The W/A timeline exercises steering. Food routing uses a separate bounded driver over public observations. Desktop execution selects an admitted Vulkan adapter at 1280×720, DPI 1.

Task	Driver	Completion / state	Bound
launch	generic.scene-load	scene-ready-and-presented-frame	120 frames
os-input	generic.desktop-keyboard	OS-key-observed-at-shared-ingress-and-public-input	120 frames
steer	generic.timeline	visible-head-turn	120 frames
eat-and-grow	test.grid-food-route-v1	food-reached-and-growth-manifests	2400 frames
self-contact	test.grid-self-contact-v1	expected-game-over-log-and-state	600 frames
reset	generic.scene-reload-or-fresh-worker	initial-scene-then-movement	180 frames
explore	generic.interactive-recording	separate-exploratory-artifact-required	120 wall s

Admission prerequisites

ID	State	Scope
runtime.public-managed-components	planned	semantic-and-game
runtime.legacy-frame-input	planned	semantic-and-game
content.native-graph-import	planned	content-and-game
render.orthographic-sorted-sprites	planned	visual-and-game
content.snake.sprite	missing	original-content-visual-and-playable
toolchain.dotnet8	check-at-run	managed
platform.isolated-linux-x11-vulkan	check-at-run	desktop-replay

Missing sprite GUID `311925a002f4447b3a28927169b83ea6`, fileID `7482667652216324306`, blocks original-content visuals. The material translation also requires implementation and validation under its selected fidelity profile.

Observe public CurrentFoodPos, active SpriteRenderer positions/enabled state, derived active segment count and the Game Over log. Validate clone-parent selectors. Assert original bytes, input-correlated state/pixels, later food/growth, reload and renewed movement, with a linked exploration record.

Captures: initial: recognizable-initial-state; turned: keyboard-to-visible-turn; grown: food-and-segment-change; reset: continued-control-after-reset. Outputs: authoritative-run.json; case-results.json; junit.xml; events.ndjson; inputs.ndjson; logs.ndjson; observations.ndjson; captures/*.png; artifact-index.json; exploratory-recording-and-note.

Budgets: 120 wall seconds; 1 GiB resident where enforceable; 10 MiB logs; 16 captures; 256 MiB per readback; 512 MiB artifacts; two compile jobs. Drain roots/worlds, complete or cancel readbacks, recheck source bytes and close only owned windows/processes.

LOCAL PRIMARY SOURCES

Repository source references

Inline citations lead here in both editions. Each path links to the local checkout used for this build. The HTML and PDF contain all report prose; opening an implementation or reference file requires this checkout.

1. ../crates/worldrunner/src/lib.rs
2. ../crates/world-model/src/lib.rs
3. ../crates/world-runtime/src/lib.rs
4. ../crates/asset-id/src/lib.rs
5. ../crates/catalog/src/lib.rs
6. ../crates/package/src/lib.rs
7. ../crates/texture/src/lib.rs
8. ../crates/mesh/src/lib.rs
9. ../crates/material/src/lib.rs
10. ../crates/disunity-ffi/src/lib.rs
11. ../crates/disunity-runtime/src/lib.rs
12. ../crates/disunity-runtime/src/bridge.rs
13. ../csharp/Disunity/WorldRunner.cs
14. ../csharp/Disunity/GameObject.cs
15. ../csharp/Disunity/MonoBehaviour.cs
16. ../csharp/Disunity/Time.cs
17. ../crates/camera/src/lib.rs
18. ../crates/render-scene/src/view.rs
19. ../crates/render-plan/src/lib.rs
20. ../crates/render-scene/src/draw.rs
21. ../crates/render-scene/src/renderer.rs
22. ../crates/render-scene/src/offscreen.rs
23. ../crates/scene-viewer/src/app.rs
24. ../crates/script/src/host.rs
25. ../crates/script/src/world.rs
26. ../crates/disunity-ffi/tests/layout.rs
27. ../crates/disunity-ffi/tests/store.rs
28. ../crates/disunity-runtime/tests/managed.rs
29. quality.md
30. training-only-unity6-reference/index.md
31. training-only-unity6-reference/final-summary.md
32. training-only-unity6-reference/game-comparison.md
33. ../crates/xtask/src/layers.rs
34. ../csharp/Disunity/NativeLibrary.cs
35. ../crates/disunity-ffi/disunity_interface.json
36. training-only-unity6-reference/game-comparison-inventory.json
37. ../tools/generate_unity_compatibility_kit_report.py
38. ../crates/disunity-ffi/build.rs
39. ../csharp/Disunity/NativeBindings.g.cs
40. ../crates/disunity-ffi/src/abi.rs
41. ../csharp/PortableFixture/NativeBridge.cs
42. ../crates/world-runtime/src/registry.rs
43. training-only-unity6-reference/G01-object-model.md
44. training-only-unity6-reference/G02-lifecycle-time.md
45. training-only-unity6-reference/G18-editor-testing.md
46. training-only-unity6-reference/G03-math.md
47. training-only-unity6-reference/G04-transforms.md
48. training-only-unity6-reference/G06-physics2d.md
49. training-only-unity6-reference/G08-input.md
50. training-only-unity6-reference/G11-ui.md
51. training-only-unity6-reference/G05-scenes-serialization.md
52. training-only-unity6-reference/G17-jobs-ecs.md
53. training-only-unity6-reference/G10-sprites-tilemaps.md
54. training-only-unity6-reference/G14-assets.md
55. training-only-unity6-reference/G09-rendering.md
56. training-only-unity6-reference/G15-platform.md

OFFLINE PRESENTATION BUILD

Regeneration and artifact checks

From the repository root:

```
python3 tools/generate_unity_compatibility_kit_report.py
```

The build uses existing Python 3, markdown_it, /usr/bin/google-chrome, pdfinfo and pdftotext. It reads the three documentation inputs below and checks local citation targets. Embedded CSS/SVG and a restrictive content policy keep the report self-contained. Chrome prints through a fresh temporary profile with background networking, extensions and sync disabled and network hosts mapped to an unreachable address.

```
pdfinfo docs/unity-compatibility-kit-architecture.pdf
pdftotext -layout docs/unity-compatibility-kit-architecture.pdf /tmp/uck-report.txt
pdftoppm -f 1 -singlefile -scale-to 1400 -png \
  docs/unity-compatibility-kit-architecture.pdf /tmp/uck-cover
```

Validation checks 54 exact catalog identities, the 50/4 split, complete source prose/code/table text in HTML and PDF, section markers, internal navigation, local citation targets and PDF word bounds. Rasterized page inspection is a separate review step. These checks establish presentation integrity; runtime and game acceptance remain implementation work.

Exact documentation input SHA-256

unity-compatibility-kit-architecture.md

`e8518d6d016ec9bbf3895111a3a5163ed0d8db3487512afc3f6b681dc24823fe`

unity-compatibility-kit-cases.md

`5103c10450b4b63175848d35b6f8ed75beaaf9bad7c90875d94c37e142a69b5a`

unity-compatibility-kit-fixture.example.json

`fb1b8ed9dbee3c008b5279f03b9a36bb1edc5e102a42eaf0746ac6d1a726733d`